SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**
COURSE OUTCOME COVERED

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CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions: hardware supports only moderate computation.
Models available: Viola-Jones 78% (low latency), YOLO 92% (moderate latency), Deep Learning 95% (high latency).

Answer:
Moderate compute rules out Deep Learning, whose high latency demand exceeds the available hardware capacity. YOLO's moderate latency fits within moderate compute while delivering strong accuracy (92%). Viola-Jones is feasible on the hardware but its accuracy is too low for reliable city-wide surveillance.

Decision: YOLO satisfies the hardware constraint while providing the best available accuracy — recommended choice.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must prioritize safety (accuracy > 90%) and maintain response time < 50 ms.
Options: Method A – 88% accuracy, 30 ms. Method B – 93% accuracy, 70 ms. Method C – 91% accuracy, 45 ms.

Answer:
Method A fails the accuracy requirement (88% < 90%). Method B fails the latency requirement (70 ms > 50 ms). Method C is the only option that satisfies both constraints simultaneously (91% > 90% and 45 ms < 50 ms).

Decision: Method C — it is the sole option meeting both the safety-accuracy and response-time requirements.

3. Motion Analysis Decision Problem

A traffic system requires accurate motion direction detection and robustness in dense traffic.
Results: Optical Flow – accurate but noisy in dense scenes. Motion Estimation – stable but less detailed.

Answer:
Under high-density traffic conditions, robustness against noise/clutter is the dominant requirement, since noisy vectors from Optical Flow would produce unreliable direction estimates when the scene is crowded.

Decision: Motion Estimation is more suitable for high-density conditions due to its stability; a hybrid design using Optical Flow in sparse traffic and switching to Motion Estimation in dense traffic would maximize both detail and robustness.

4. Background Modeling Evaluation

A GMM-based system produces: Static scenes – 90% accuracy. Dynamic scenes – 65% accuracy.

Answer:
The 25-point accuracy drop in dynamic scenes shows GMM alone does not reliably meet requirements where background motion (e.g., trees, water, crowds) is present.

Decision: GMM alone is insufficient for environments with dynamic backgrounds. Recommend supplementing GMM with adaptive background modeling or a deep-learning-based subtraction method for dynamic regions, while retaining GMM for the static regions where it already performs well (hybrid approach).

5. Depth Estimation Strategy

Constraints: only one camera available, moderate depth accuracy acceptable.

Options: Stereo Vision – high accuracy, needs two cameras. Structure from Motion – moderate accuracy, single camera.

Answer:

Stereo Vision cannot be deployed at all given the single-camera hardware constraint, regardless of its higher accuracy.

Decision: Structure from Motion is the only feasible option, and its moderate accuracy is explicitly stated as acceptable, so it fully satisfies the given constraints.

6. Retail Motion Tracking Logic

A retail system requires accurate tracking when crowd density is low and stable tracking when density is high. Observations: Optical Flow – accurate in low density, noisy in high density. Motion Models – stable in high density, less detailed.

Answer:

Neither method alone satisfies both requirements across all density levels: Optical Flow meets the low-density accuracy need but fails the high-density stability need, and vice-versa for Motion Models.

Decision: adopt a combined/adaptive approach — use Optical Flow when real-time crowd-density estimation indicates low density, and switch to Motion Models when density is high. This satisfies both stated requirements logically.

7. Defect Detection System Decision

Requirements: detect subtle defects (>95% accuracy), adapt to product variations.

Options: Traditional methods – fast, less adaptive. Deep learning – accurate, requires training.

Answer:

Traditional methods are unlikely to reach the >95% accuracy threshold for subtle defects and lack adaptability to product variation. Deep learning can be trained/retrained to adapt to new product variants and is capable of exceeding the accuracy requirement with sufficient training data.

Decision: Deep Learning is recommended, since it directly satisfies both the accuracy and adaptability requirements, despite its added training overhead.

8. AR Motion Tracking Constraint Problem

An AR system must maintain stable tracking (<5% drift) and run on mobile hardware.

Options: Optical Flow – detailed but computationally intensive. Motion Estimation – less detailed but efficient.

Answer:

Mobile hardware has limited compute/power, so Optical Flow's high computational load risks not running smoothly on-device, even though it is more detailed. Motion Estimation's efficiency fits mobile hardware constraints directly.

Decision: Motion Estimation is recommended for the mobile hardware constraint; if the resulting drift approaches or exceeds 5%, periodic keyframe correction using lightweight Optical Flow snapshots can be added to keep drift within bounds without breaching compute limits.

9. Mobile Face Detection Trade-off

Constraints: real-time operation required.

Options: Viola-Jones – 80%, fast. Deep Learning – 95%, slow.

Answer:

Real-time operation is a hard constraint, and Deep Learning's slow inference is likely to violate it on typical mobile hardware, despite its higher accuracy.

Decision: Viola-Jones is recommended for strict real-time mobile operation. (Note: if the mobile hardware includes GPU/NPU acceleration, a lightweight deep model such as MobileFaceNet could bridge the accuracy gap while remaining real-time — but of the two given options, Viola-Jones satisfies the stated constraint.)

10. Animation System Optimization

A game engine requires realistic motion for cutscenes and fast rendering for gameplay.

Options: Motion Models – realistic, computationally heavy. Motion Estimation – efficient, less realistic.

Answer:

The two requirements (cinematic realism vs. real-time gameplay speed) are contradictory for a single technique, so no single method satisfies both contexts.

Decision: design the system to use Motion Models during pre-rendered/offline cutscenes where computation budget is flexible, and switch to Motion Estimation during live gameplay rendering where speed is critical — a context-dependent hybrid design.

11. Traffic Surveillance Model Selection

A traffic monitoring system must work under varying lighting.

Options: Method A – 88%, 60 ms. Method B – 91%, 85 ms. Method C – 92%, 75 ms.

Answer:

No explicit latency ceiling is given, so accuracy under varying lighting is the primary differentiator. Method C offers the highest accuracy (92%) at a moderate latency (75 ms), better than Method B's latency (85 ms) for less accuracy, and clearly ahead of Method A on accuracy.

Decision: Method C offers the best overall balance of accuracy and response time and is recommended.

12. Drone Vision System Constraint

A drone system requires lightweight computation and real-time detection (<60 ms).

Options: Model A – 83%, 40 ms. Model B – 87%, 65 ms. Model C – 86%, 55 ms.

Answer:

Model B fails the <60 ms constraint (65 ms). Between the remaining options, Model C (55 ms, 86%) offers higher accuracy than Model A (40 ms, 83%) while still satisfying the real-time requirement.

Decision: Model C is recommended — it meets the <60 ms latency constraint with the best accuracy among the compliant options.

13. Security System Recognition

A security system must have high accuracy (>92%) and moderate computation.

Options: Traditional – 85%, low computation. Deep Learning – 94%, high computation. Hybrid – 91%, moderate computation.

Answer:

Traditional fails the accuracy requirement. Deep Learning meets the accuracy requirement (94% > 92%) but fails the moderate-computation constraint. Hybrid satisfies the computation constraint but falls just short of the accuracy threshold (91% < 92%).

Decision: since no option satisfies both constraints exactly, the choice depends on priority — if accuracy is non-negotiable for security, select Deep Learning and provision additional compute resources; if computation budget is a hard limit, Hybrid is the closest practical compromise, accepting a 1-point accuracy shortfall.

14. Motion Tracking in Sports Analytics

Requirements: accurate motion tracking, real-time performance.

Options: Optical Flow – accurate, slow. Motion Estimation – fast, less accurate.

Answer:

For live broadcast overlays requiring strict real-time output, Motion Estimation's speed is essential even at reduced accuracy. For post-event or slightly delayed analytics where a small buffer is tolerable, Optical Flow's higher accuracy is more valuable.

Decision: use Motion Estimation for live/real-time broadcast tracking, and Optical Flow for near-real-time detailed performance analysis where a short processing delay is acceptable.

15. Face Recognition System Design

Constraints: works in low light, moderate computation.

Options: Traditional – 82%. Deep Learning – 93%. Hybrid – 89%.

Answer:

Deep Learning likely requires high computation, exceeding the moderate-computation constraint, while Traditional's low accuracy may be inadequate in low light. Hybrid balances reasonable accuracy (89%) with moderate computational demand, aligning with the stated constraint.

Decision: Hybrid is recommended as it best satisfies the moderate-computation requirement while maintaining acceptable accuracy for low-light conditions.

16. Autonomous Navigation Constraint

System must have latency < 70 ms.

Options: Model A – 94%, 60 ms. Model B – 96%, 80 ms. Model C – 95%, 65 ms.

Answer:

Model B fails the latency constraint (80 ms > 70 ms). Both Model A (60 ms) and Model C (65 ms) satisfy the constraint; Model C has higher accuracy (95% vs 94%) while still meeting the latency requirement.

Decision: Model C is recommended — it offers the best accuracy among the models that satisfy the <70 ms constraint.

17. Industrial Inspection System

Requirements: real-time operation.

Options: Method A – 92%, fast. Method B – 96%, slow. Method C – 95%, moderate.

Answer:

Method B's slow speed likely conflicts with real-time inspection line requirements. Method A is fastest but has the lowest accuracy of the three; Method C offers near-Method-B accuracy (95%) at moderate speed, which can typically still satisfy real-time inspection throughput.

Decision: Method C is recommended as the best balance of accuracy and real-time feasibility; if the production line demands the fastest possible cycle time, Method A is the fallback choice.

18. Crowd Monitoring System

Requirements: stable detection in dense crowds, moderate accuracy acceptable.

Options: Optical Flow – unstable in dense scenes. Motion Models – stable.

Answer:

Optical Flow directly fails the stability requirement in dense crowds. Motion Models satisfy the stability requirement, and since only moderate accuracy is required, its lower level of detail is acceptable.

Decision: Motion Models is the clear choice, fully satisfying the stated constraint.

19. Video Analytics Pipeline

Constraints: processing time < 100 ms.

Options: Pipeline A – 88%, 80 ms. Pipeline B – 91%, 120 ms. Pipeline C – 90%, 95 ms.

Answer:

Pipeline B fails the <100 ms constraint. Between A (80 ms, 88%) and C (95 ms, 90%), both satisfy the latency constraint; Pipeline C offers higher accuracy while remaining within the threshold, though with a smaller safety margin.

Decision: Pipeline C is recommended for the accuracy gain; Pipeline A is the safer choice if consistent processing-time variance near the 100 ms limit is a concern.

20. Smart Parking System

Requirements: detect vehicles accurately, low computational cost.

Options: Viola-Jones – low cost, low accuracy. YOLO – moderate cost, high accuracy.

Answer:

The requirement explicitly prioritizes accurate detection. Viola-Jones's low accuracy risks missed or incorrect vehicle detection, undermining the system's core purpose, even though it is cheaper computationally.

Decision: YOLO is recommended since it satisfies the explicit accuracy requirement at a computational cost that is only moderate, not prohibitive; Viola-Jones is reserved for extremely resource-constrained deployments where some accuracy loss is tolerable.

21. Medical Imaging Analysis

Requirements: very high accuracy (>95%), processing time not critical.

Options: Traditional – 85%. Deep Learning – 97%.

Answer:

Traditional fails the accuracy requirement (85% < 95%). Deep Learning meets it (97% > 95%), and since processing time is explicitly stated as not critical, its higher computational cost is not a limiting factor.

Decision: Deep Learning is the clear and unambiguous choice, as it is the only option satisfying the accuracy requirement and the time constraint is not binding.

22. Edge Device Vision System

Constraints: low power, real-time performance.

Options: Deep Learning – accurate, heavy. Lightweight Model – moderate accuracy, efficient.

Answer:

Deep Learning's heavy computation conflicts directly with the low-power constraint of edge devices. The Lightweight Model's efficiency satisfies both the low-power and real-time requirements.

Decision: Lightweight Model is recommended for edge deployment, accepting moderate accuracy in exchange for meeting the hard power and real-time constraints.

23. Traffic Flow Analysis

Requirements: detect motion patterns, robust in crowded scenes.

Options: Optical Flow – detailed, noisy. Motion Estimation – stable.

Answer:

Optical Flow's noise in crowded scenes directly conflicts with the robustness requirement. Motion Estimation's stability satisfies the robustness requirement even though it provides less fine-grained detail.

Decision: Motion Estimation is recommended, since robustness in crowded scenes is the binding requirement here.

24. Retail Analytics System

Requirements: customer tracking, moderate accuracy, real-time processing.

Options: Method A – accurate, slow. Method B – moderate accuracy, fast.

Answer:

Method A's slow speed conflicts with the real-time processing requirement, regardless of its higher accuracy. Method B satisfies both the real-time and moderate-accuracy requirements as explicitly stated.

Decision: Method B is recommended, since it is the only option meeting both stated constraints.

25. Object Detection in Foggy Conditions

Requirements: robust under poor visibility.

Options: Model A – 80%. Model B – 87%. Model C – 90% but slow.

Answer:

If real-time operation is not explicitly required here, Model C's higher accuracy (90%) makes it preferable for safety-critical, visibility-limited scenarios such as fog, despite the added latency.

Decision: recommend Model C when latency can be tolerated (e.g., driver-assist warnings with buffer time); if strict real-time response is required, Model B (87%) is the best balance of accuracy and speed among the remaining options.

26. Video Surveillance Upgrade

Constraints: improve accuracy, maintain same latency.

Options: Model A – 85%, 50 ms. Model B – 90%, 50 ms.

Answer:

Model B improves accuracy by 5 points over Model A while keeping the exact same latency (50 ms), directly satisfying both stated constraints.

Decision: Model B is recommended, as it meets the explicit requirement of improved accuracy at unchanged latency with no trade-off.

27. Robotics Vision System

Requirements: fast decision-making, moderate accuracy.

Options: Deep Learning – slow. Traditional – fast.

Answer:

Deep Learning's slow inference directly conflicts with the fast decision-making requirement for robotics control loops. Traditional's speed satisfies this hard constraint, and the accuracy requirement is only moderate, which Traditional methods can typically meet.

Decision: Traditional method is recommended for robotics requiring fast decision cycles.

28. Smart Home Security

Requirements: detect intrusions accurately, low computation.

Options: YOLO – accurate. Viola-Jones – efficient.

Answer:

Accurate intrusion detection is explicitly prioritized, and YOLO better satisfies this, though it typically demands more compute than Viola-Jones.

Decision: if the device has at least minimal acceleration (e.g., a lightweight YOLO variant on an edge chip), YOLO is recommended for its accuracy; for extremely constrained low-power hardware where compute is a hard ceiling, Viola-Jones is the necessary fallback despite lower accuracy.

29. Autonomous Drone Tracking

Constraints: real-time (<40 ms).

Options: Model A – 83%, 30 ms. Model B – 86%, 50 ms. Model C – 88%, 38 ms.

Answer:

Model B fails the <40 ms constraint (50 ms). Between Model A (30 ms, 83%) and Model C (38 ms, 88%), both satisfy the latency constraint, and Model C offers notably higher accuracy while still meeting it.

Decision: Model C is recommended, since it delivers the best accuracy among the models that satisfy the real-time constraint.

30. Motion Animation System

Requirements: smooth motion, real-time rendering.

Options: Motion Models – realistic, slow. Motion Estimation – fast, less realistic.

Answer:

Motion Models' slow processing directly conflicts with the real-time rendering requirement, regardless of its superior realism. Motion Estimation's speed satisfies the hard real-time constraint.

Decision: Motion Estimation is recommended for the real-time rendering pipeline; Motion Models can still be used for any pre-rendered/offline segments (e.g., cutscenes) where real-time rendering is not required, mirroring the hybrid approach used in Q10.

Code of Conduct:

I V M SAI BHARGAV (192324126) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V M SAI BHARGAV